



A VIBE CODED LASER STABILIZATION PID SYSTEM

High precision. Real-time control. Effortless stability.



High Precision

Stable laser output
with low noise



Real-time Control

Vibe coded PID for
fast & accurate response



Smart Monitoring

Intuitive dashboard
for live insights



Divalent atoms : opportunities

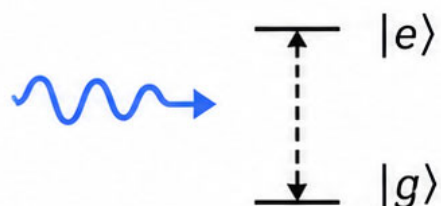


中央研究院
ACADEMIA SINICA

A CONTROL

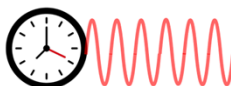
Qubit control

Coherent manipulation of atomic states



- ✓ Prepare superposition states
- ✓ Drive transitions
- ✓ Implement quantum gates

Coherence Time

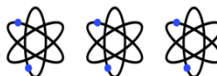


Quantum Properties

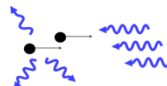


$$|g\rangle + e^{i\phi}|e\rangle$$

Identity



Scalability



Qubits

hyperfine state qubit

$$^3P_2 = |m\rangle$$

$$^3P_0 = |m\rangle$$

Optical qubit

$$^1S_0 = |g\rangle$$

3D manifold

Telecom wavelength (flying qubit)

$$|1\rangle$$

$$|0\rangle$$

metastable state qubit

$$|1\rangle$$

$$|0\rangle$$

Ground state qubit

Divalent atoms : opportunities

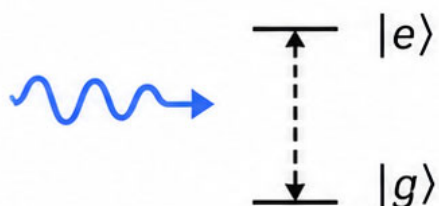


中央研究院
ACADEMIA SINICA

A CONTROL

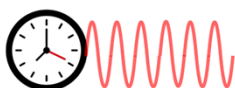
Qubit control

Coherent manipulation of atomic states

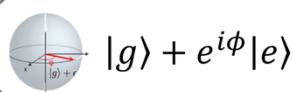


- ✓ Prepare superposition states
- ✓ Drive transitions
- ✓ Implement quantum gates

Coherence Time



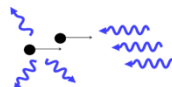
Quantum Properties



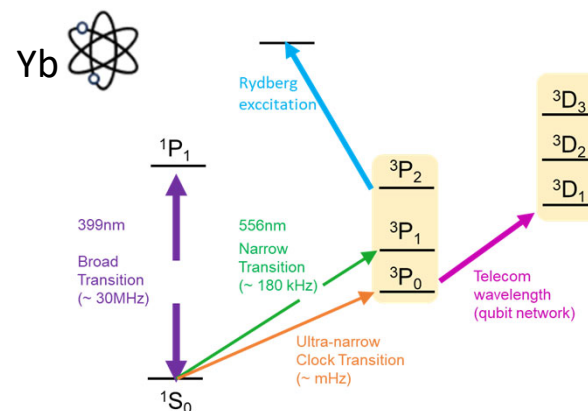
Identity



Scalability



Atomic system



Divalent atoms : **tools**

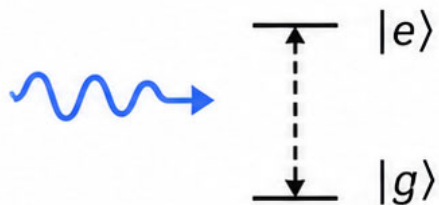


中央研究院
ACADEMIA SINICA

A CONTROL

Qubit control

Coherent manipulation of atomic states



- ✓ Prepare superposition states
- ✓ Drive transitions
- ✓ Implement quantum gates

MAGNETIC FIELD

Define quantization axis and lift degeneracy



LIGHT

Coherent excitation and readout



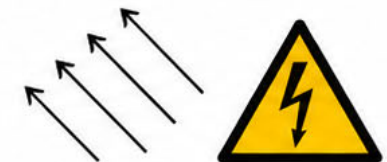
- ✓ Drive transitions (laser cooling)
- ✓ State preparation
- ✓ Fluorescence readout

ATOM

Divalent atom



ELECTRIC FIELD



- ✓ AC - Stark shift
- ✓ DC - Stark shift



Goal: Precisely control and measure atoms to explore quantum phenomena and build quantum technologies.

Quantum engineer



中央研究院
ACADEMIA SINICA

Bridging quantum physics, engineering, and AI to build the technologies of tomorrow.



March 24, 2026



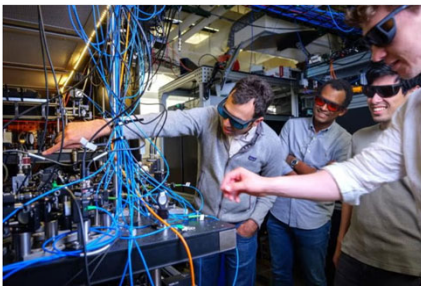
Mountain View, California

JILA Joint Institute for Laboratory Astrophysics

About Research Centers People News Outreach Jobs Events

Search

Google Quantum AI Engages JILA Fellow Adam Kaufman to Lead New Neutral Atom Quantum Computing Effort



Google has engaged theoretical atomic physicist and JILA Fellow Adam Kaufman to lead a new quantum computing effort using neutral atoms.

Kaufman, an internationally recognized leader in ultracold physics, will continue his research at JILA while working with Google Quantum AI to advance neutral atom quantum computers—promising platforms for solving problems beyond the reach of today's supercomputers.



Goal: Advance neutral atom quantum computing by combining fundamental science, engineering excellence, and AI-driven innovation.



Research Scientist, Neutral Atoms, Quantum AI

Google Boulder, CO, USA M5

Minimum qualifications

- PhD degree in Physics or Electrical Engineering, or equivalent practical experience.
- One or more scientific publication submissions for conferences, journals, or public repositories related to quantum.

[Learn more](#)



Staff Research Scientist Manager, Neutral Atoms, Quantum AI

Google Boulder, CO, USA Advanced

Minimum qualifications

- PhD degree in Physics or Electrical Engineering, or equivalent practical experience.
- 7+ years of experience managing a team.
- One or more scientific publication submission(s) for conferences, journals, or public repositories related to quantum.

[Learn more](#)



Research Scientist, Optical Engineering Neutral Atoms, Quantum AI

Google Boulder, CO, USA M4

Minimum qualifications

- PhD in Optics, Physics, Electrical Engineering, or equivalent practical experience.
- Experience in the design, implementation, and characterization of optical systems.
- One or more scientific contributions, including publications in peer-reviewed journals or presentations at quantum technology conferences.

[Learn more](#)

Quantum engineer



中央研究院
ACADEMIA SINICA

Bridging quantum physics, engineering, and AI to build the technologies of tomorrow.



March 24, 2026



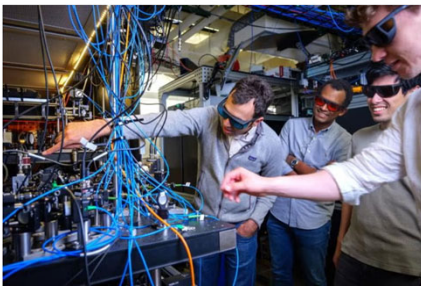
Mountain View, California

JILA Joint Institute for Laboratory Astrophysics

About Research Centers People News Outreach Jobs Events

Search

Google Quantum AI Engages JILA Fellow Adam Kaufman to Lead New Neutral Atom Quantum Computing Effort



Google has engaged theoretical atomic physicist and JILA Fellow Adam Kaufman to lead a new quantum computing effort using neutral atoms.

Kaufman, an internationally recognized leader in ultracold physics, will continue his research at JILA while working with Google Quantum AI to advance neutral atom quantum computers—promising platforms for solving problems beyond the reach of today's supercomputers.



Goal: Advance neutral atom quantum computing by combining fundamental science, engineering excellence, and AI-driven innovation.



Research Scientist, Neutral Atoms, Quantum AI

Google Boulder, CO, USA M5

Minimum qualifications

- PhD degree in Physics or Electrical Engineering, or equivalent practical experience.
- One or more scientific publication submissions for conferences, journals, or public repositories related to quantum.

[Learn more](#)



Staff Research Scientist Manager, Neutral Atoms, Quantum AI

Google Boulder, CO, USA Advanced

Minimum qualifications

- PhD degree in Physics or Electrical Engineering, or equivalent practical experience.
- 7+ years of experience managing a team.
- One or more scientific publication submission(s) for conferences, journals, or public repositories related to quantum.

[Learn more](#)



Research Scientist, Optical Engineering Neutral Atoms, Quantum AI

Google Boulder, CO, USA M4

Minimum qualifications

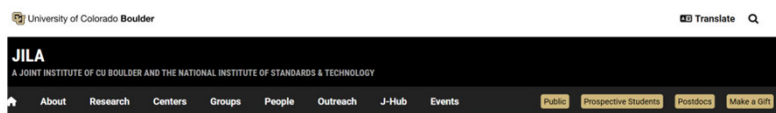
- PhD in Optics, Physics, Electrical Engineering, or equivalent practical experience.
- Experience in the design, implementation, and characterization of optical systems.
- One or more scientific contributions, including publications in peer-reviewed journals or presentations at quantum technology conferences.

[Learn more](#)

Quantum engineer

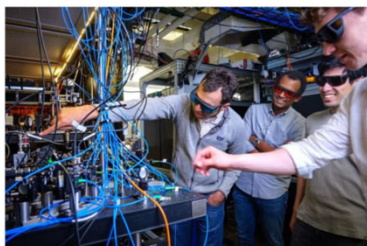


3/24/2026



Google Quantum AI Engages JILA Fellow Adam Kaufman to Lead New Neutral Atom Quantum Computing Effort

3/24/2026 · By Steven Burrows / JILA Science Communications Manager



Adam Kaufman (left) inspects an optical atomic clock at JILA on the University of Colorado campus with students Nelson Darkwah Oppong, Alec Cao and Theo Lukin Yelin. (Image Credit: Patrick Campbell, University of Colorado)

Today, Google Quantum AI announced a major expansion of its quantum computing research program, naming JILA Fellow Adam Kaufman to lead a newly formed neutral atom quantum hardware team. The initiative marks Google's first large-scale investment in neutral atom quantum computing, a rapidly advancing platform that complements its long-standing work in superconducting qubits.

Kaufman, an internationally recognized leader in neutral atom physics, will continue his research at JILA as a JILA Fellow while maintaining his academic appointment in the Department of Physics at the University of Colorado Boulder. According to The Colorado Sun, Kaufman's dual role reflects Google's strategy of closely integrating industrial-scale engineering with cutting-edge academic research, particularly in Boulder's growing quantum ecosystem.

In announcing the program, Google emphasized that neutral atom quantum processors offer unique advantages, including the ability to scale to very large arrays—currently on the order of thousands to tens of thousands of qubits—with highly flexible, “any-to-any” connectivity. While neutral atom systems operate more slowly than superconducting circuits, their scalability in qubit number makes them especially promising for quantum simulation and fault-tolerant architectures. Google views the parallel development of both platforms as a way to accelerate progress toward commercially useful quantum computers.

財經 > 證券產業

廣達強化佈局量子電腦 砸下8.9億投資法國新創Pasqal

2026/01/28 17:54



廣達電腦股份有限公司

職務類別 光電工程研發主管、硬體工程研發主管、其他工程研發主管

工作性質 全職

上班地點 桃園市龜山區文化二路211號 (林口華亞科技園區)

原分子光學工程師(AMO Engineer) - J2432

待遇面議 (經常性薪資達 4 萬元或以上) ?

03/12

我要應徵

工作內容

1. 設計並操作冷原子、離子阱等系統實驗平台。
2. 跨單位合作設計高穩定性雷射系統及螢光量測系統，並整合光學元件進行測試。
3. 撰寫控制程式與資料分析工具，提升系統穩定性與



關注



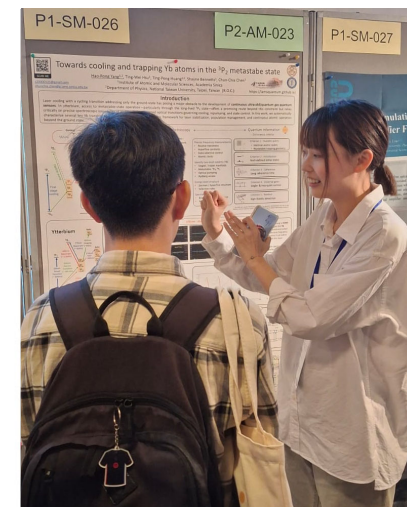
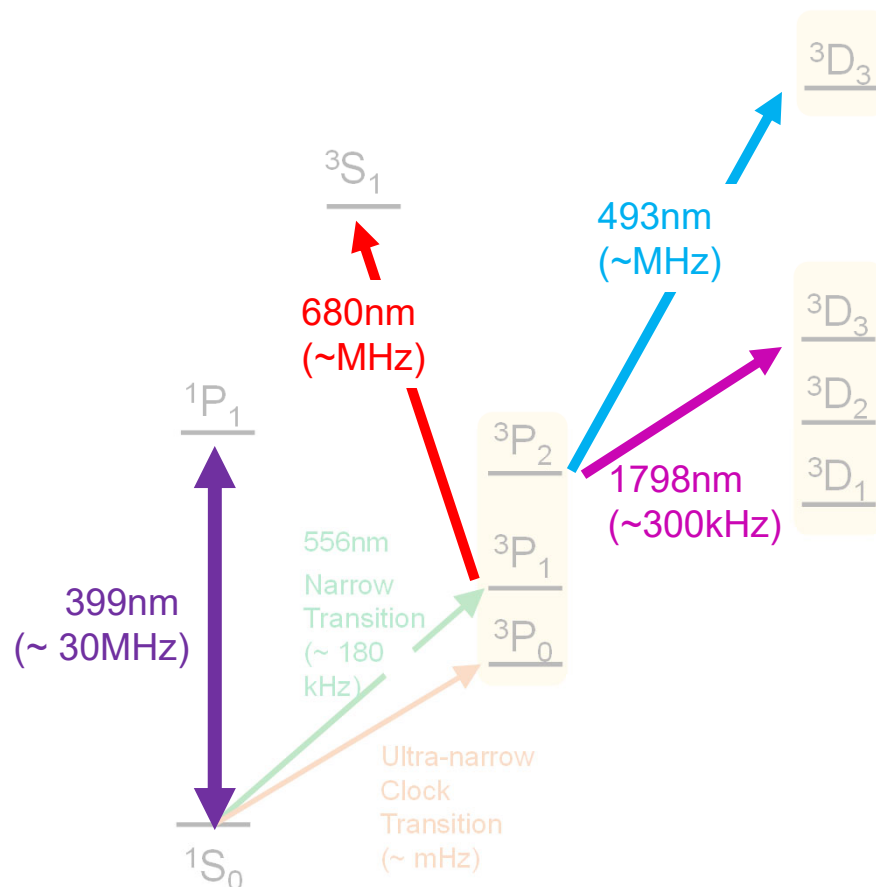
儲存

我要應徵

Quantum engineer ready ??



中央研究院
ACADEMIA SINICA

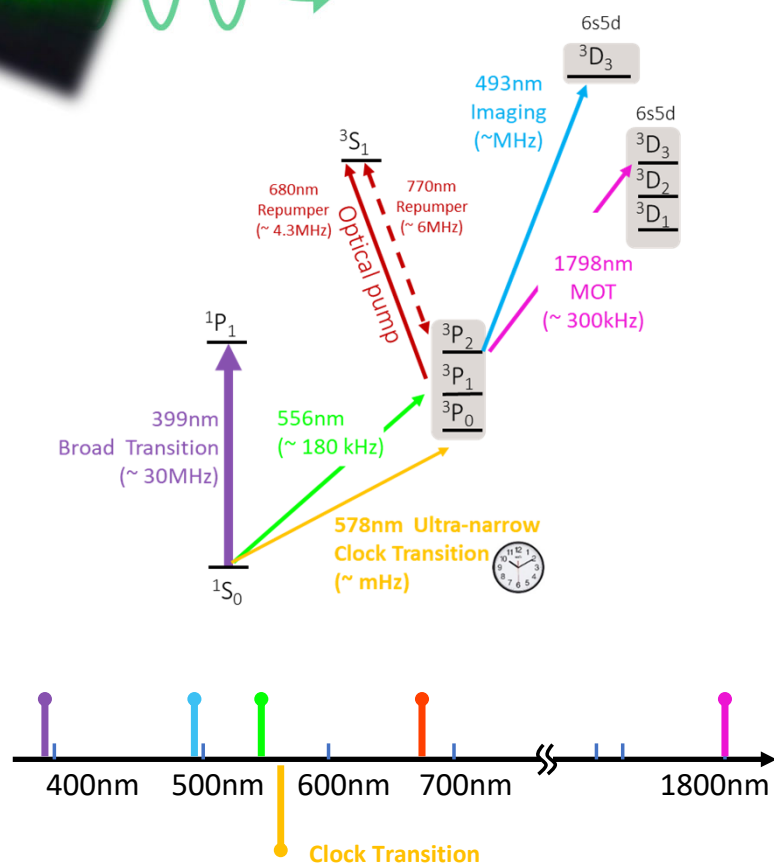
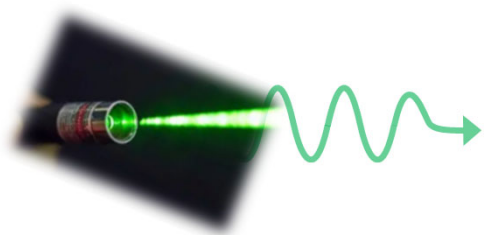


Hao-Rong Yang 楊皓蓉
Master thesis (in preparation)

Meter (voltmeter vs. wavemeter)



中央研究院
ACADEMIA SINICA



(A) HighFinesse_Wavelength Meters
with IR-I Range (630-1750nm)



(B) Moglabs (350 - 1350nm)



Meter (voltmeter vs. wavemeter)

Masterhuang

水電爸爸的 水電DIY 實務課

居家水電自己來！

水電爸爸

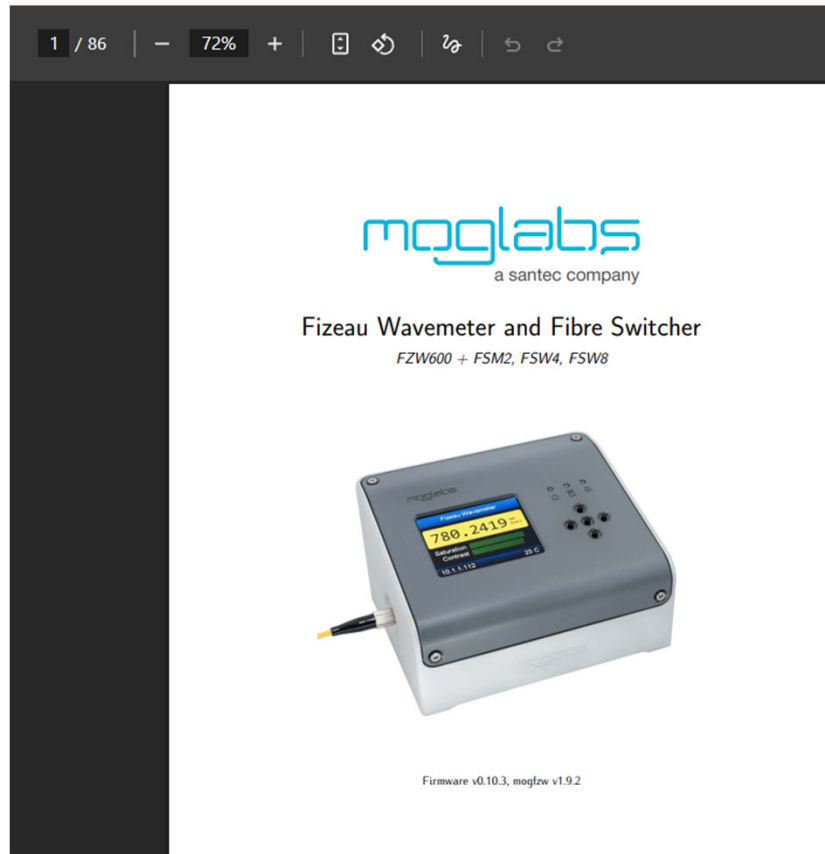
簡單好懂、白話翻譯的水電實作技巧
讓水電媽媽帶你一起升級家中每個空間！

水電媽媽 | 黃筠

擅長老屋翻新、開店裝修，一手包辦水電規劃與現場施工，協助許多家庭打造理想家居

- 行鈞流水電工程創辦人
- 政府水電職訓班特聘講師
- FB「水電媽媽」累積超過 9.2 萬追蹤
- 具備 7 項專業資格：
甲級電器承裝業、乙級自來水管承裝業、
乙級室內配線、乙級家具木工...等





> Command language

The FZW can be controlled over USB via a virtual serial port, or over Ethernet using TCP/IP. The syntax follows a text-based request/reply architecture with messages delimited by CRLF. Failed queries are replied to with the string "ERR" followed by an explanation of the issue. It is strongly recommended to check for the response before sending the next command.

B.3 Measurement

AVERAGE `MEAS,AVERAGE[,val]`

Specify the wavelength measurement to average over `val` milliseconds for improved measurement precision at the expense of measurement rate. `val` is limited to 60,000 (1 minute averaging time). If `val` is zero, no averaging is performed. If used with a fibre switcher, the averaging time can be different for each channel. Note: behaviour is unknown if used with auto-stepping of a fibre switcher.

CLEAR `MEAS,CLEAR`

Reset the measurement averaging and internal verification, for use in combination with an external optical switch or shutter.

CONTRAST `MEAS,CONTRAST`

Returns the measurement contrast, which is a number in the range [0, 100] that measures the fringe quality.

CORRECT `MEAS,CORRECT,val`

Apply a correction to the device internal calibration using the currently-supplied laser as an absolute reference. The reference `val` should be specified *in THz* and as many significant figures as practical. Perturbing the calibration in this way will improve the absolute accuracy of the device around the reference wavelength, at the expense of the absolute accuracy far from the reference wavelength.

Lazy prompt



中央研究院
ACADEMIA SINICA



Hi, is there a easy way to use python to control the frequency of the laser via moglab wavemeter

Moglab wavemeter

Chat

- New chat
- Projects
- Artifacts
- Customize

Recents

- Quantum metrology beat...
- Exponential motion appro...
- Moglab wavemeter
- S13 CSV processing windo...
- Develop performance ben...

Yes! The MOGLabs FZW has a straightforward text-based command interface over TCP/IP or USB, so Python control is quite simple. Here's a minimal example to get you started:

```
python
import socket
import time

class FZW:
    def __init__(self, ip, port=7802):
        self.sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        self.sock.connect((ip, port))
        self.sock.settimeout(2.0)

    def send(self, cmd):
        self.sock.sendall((cmd + '\r\n').encode())
        response = self.sock.recv(1024).decode().strip()
        return response
```

Moglab wavemeter

i want to design an interface that I can do the frequency control of the laser, on the GUI interface, i would like to configure the laser PID to control the laser wavelength. a similar feature like highFinesse wavemeter

1 Apr

Let me build you a professional laser frequency control GUI similar to the HighFinesse interface shown.

Laser Frequency Controller PID OFF Unlocked Disconnected

Current frequency —	Saturation —	Contrast —
Error from setpoint — MHz	PID output — V	Temperature — °C

Reference PID settings Exposure Connection

Setpoint (THz) 384.229242

Setpoint (nm vac) 780.2419

Course signal f(t) constant

Use latest measurement THz → nm

Enable PID Single shot Clear history Live update

Frequency vs time Span (GHz) 0.02 Duration (s) 60

Mean: — Std: — Range: —

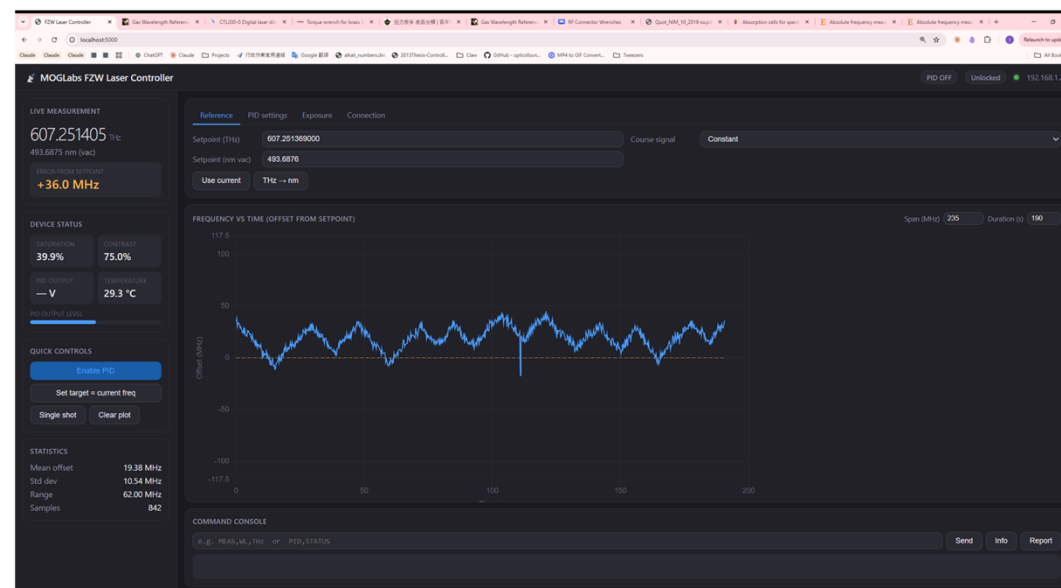
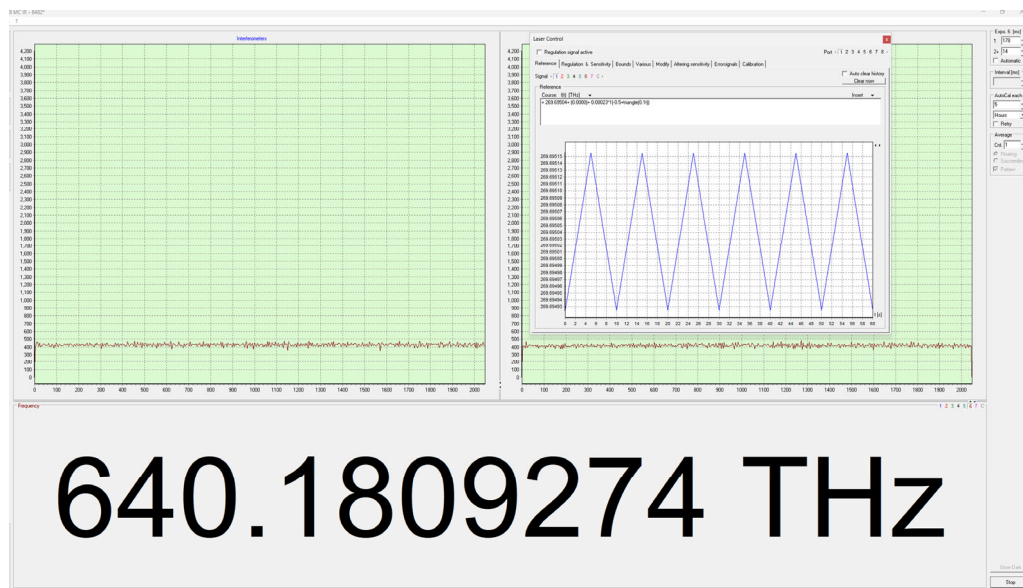
Command: e.g. MEAS,WL,THz Send



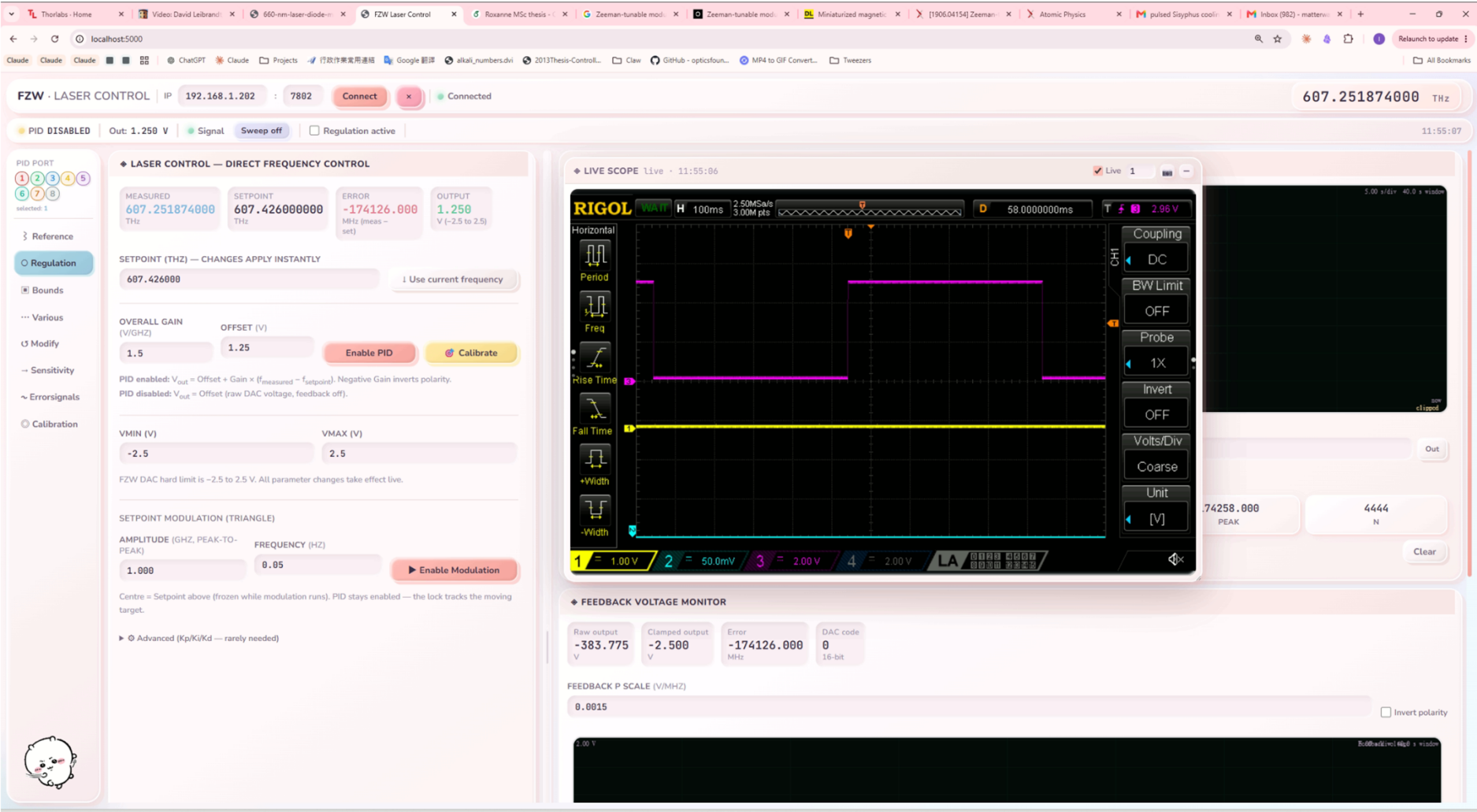
中央研究院
ACADEMIA SINICA



VS.



Current status





◆ LASER CONTROL — DIRECT FREQUENCY CONTROL

MEASURED

607.250927000

THz

SETPOINT

607.250000000

THz

ERROR

+41.535

MHz (meas – set)

OUTPUT

1.370

V (–2.5 to 2.5)

SETPOINT (THz) — CHANGES APPLY INSTANTLY

607.250

Live: 607.25080492 THz

Use current frequency

OVERALL GAIN (V/GHZ)

1.5

OFFSET (V)

1.25

Disable PID

Calibrate

PID enabled: $V_{out} = \text{Offset} + \text{Gain} \times (f_{measured} - f_{setpoint})$. Negative Gain inverts polarity.

PID disabled: $V_{out} = \text{Offset}$ (raw DAC voltage, feedback off).

VMIN (V)

–2.5

VMAX (V)

2.5

FZW DAC hard limit is –2.5 to 2.5 V. All parameter changes take effect live.

SETPOINT MODULATION (TRIANGLE)

AMPLITUDE (GHZ, PEAK-TO-PEAK)

2

FREQUENCY (HZ)

0.01

Disable Modulation

Centre = Setpoint above (frozen while modulation runs). PID stays enabled — the lock tracks the moving target.

▼ Advanced (Kp/Ki/Kd — rarely needed)

KP

0.9

KI

0.3

KD

0

FZW time-series - A03226

Plot: THz

☐ Symbols

☐ Indicate errors

Centre: 607.250000 THz

Span: 6.000 GHz

☐ Auto range

Measurement: Duration: 500.0 s Interval: 0.000 s

☐ Averaging

☐ Scrolling

Pause

Single shot

Clear

Export image

Export data

Frequency (THz)

607.253

607.252

607.251

607.250

607.249

607.248

607.247

0

100

200

300

400

500

Time (s)

Mean:

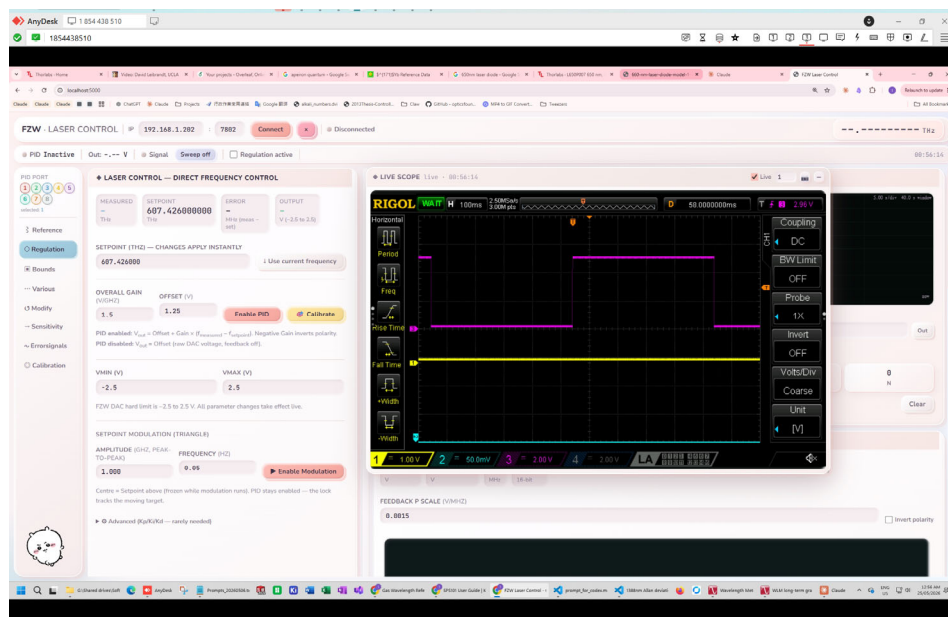
607.249980 THz

65737 vals, 146.7 meas/s

Mean: 607.249980 THz

Std. Dev: 599.2 MHz

Range: 2.0 GHz



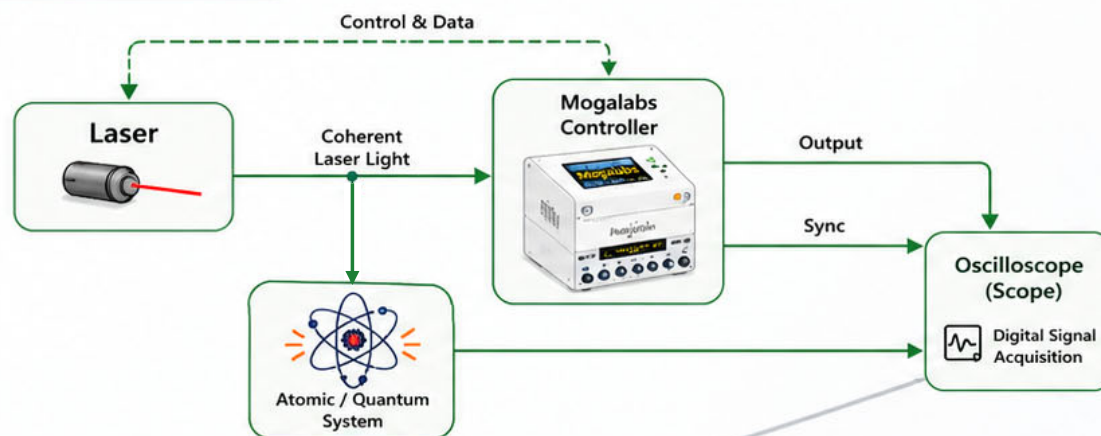
- Rigol DS1000Z / DS1104Z (raw SCPI over TCP)
- Default address: 192.168.1.168 : 5555.
- Commands terminated \n. Text queries read until \n. Binary queries return IEEE 488.2 definite-length blocks: #N<NN...N><bytes>\n.
- Commands the server uses:
 - *IDN?,
 - :CHAN<n>:SCAL <v>,
 - :TIM:SCAL <t>,
 - :CHAN<n>:DISP ON|OFF,
 - :WAV:SOUR CHAN<n>,
 - :WAV:MODE NORM,
 - :WAV:FORM BYTE,
 - :WAV:PRE?,
 - :WAV:DATA?,
 - :DISP:DATA? ON,OFF,BMP24.

Tool: Laser control and data Acquisition system

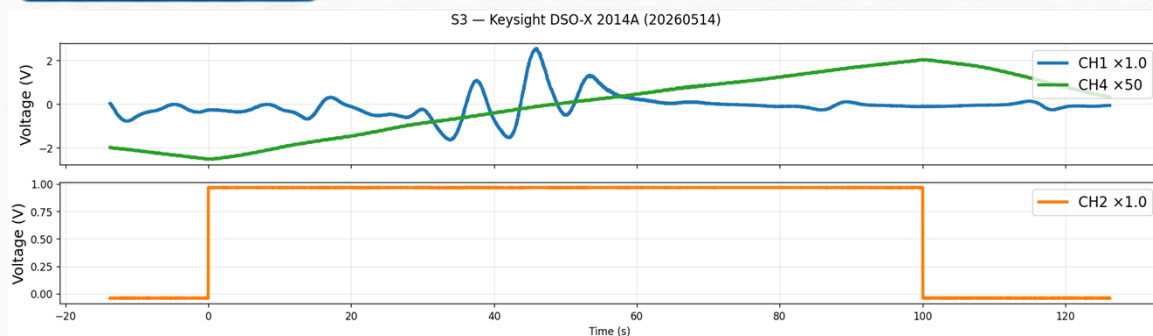


中央研究院
ACADEMIA SINICA

System Architecture



Real-time Monitoring



Integrated Control

Unified control of laser, modulation, and acquisition.



Real-time Visualization

Live I/Q signals and digital outputs for fast diagnostics.



Advanced Modulation

Flexible modulation (PD) with feedback and lock-in support.

Laser Control — Direct Frequency Control

MEASURED

607.259927000
THz

SETPOINT

607.250000000
THz

ERROR

+41.535
MHz (Prev: -)
MHz

OUTPUT

1.379
V (± 5.0 V, 2.5 Ω)

SETPOINT (THz) — CHANGES APPLY IMMEDIATELY

607.250 THz

Use current frequency

Limit: 407–239880000 THz

OVERALL GAIN (Kp)

1.5

OFFSET (V)

0.25

Disable PID

Calibrate

PID enabled: $V_{out} = \text{Offset} + \text{Gain} \times (f_{\text{setpoint}} - f_{\text{measured}})$. Negative Gain inverts polarity.

PID disabled: $V_{out} = \text{Offset}$ (use DAC settings: feedback off)

VMAX (V)

+2.5

VMAX (V)

2.5

FW VOC head limit: ± 2.5 to ± 2.5 V. All parameter changes take effect live.

SETPOINT MODULATION (TRIANGLE)

AMPLITUDE (p-p) (0–2.4 TO VMAX)

2

FREQUENCY (Hz)

0.01

Disable Modulation

Center = Setpoint above (mean while modulation runs). PD stays enabled — the lock tracks the moving target.

▼ Q Advanced (Kp, Ki, Kd — rarely needed)

KP

0.9

KI

0.3

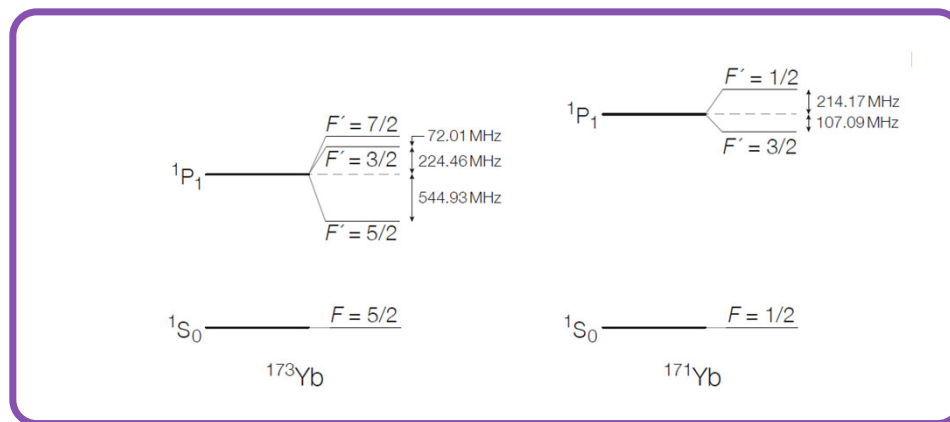
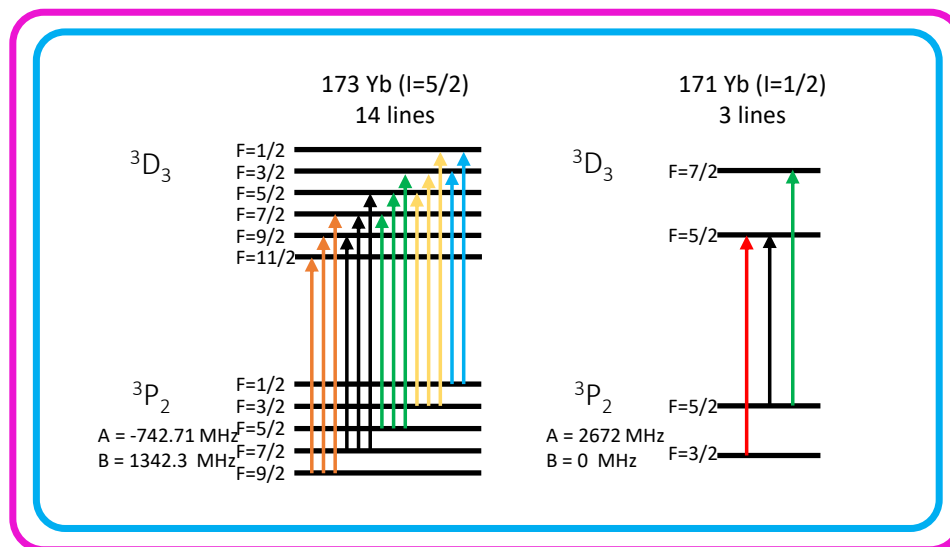
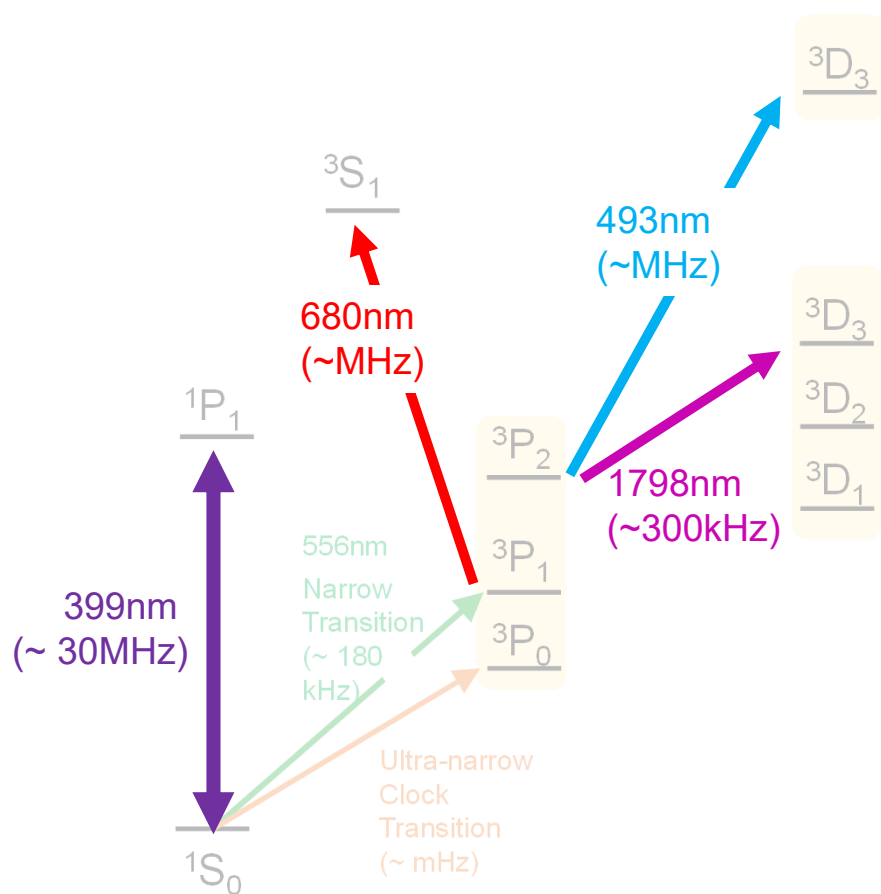
KD

0

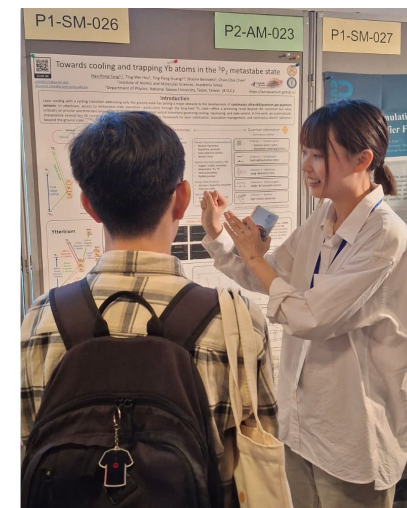
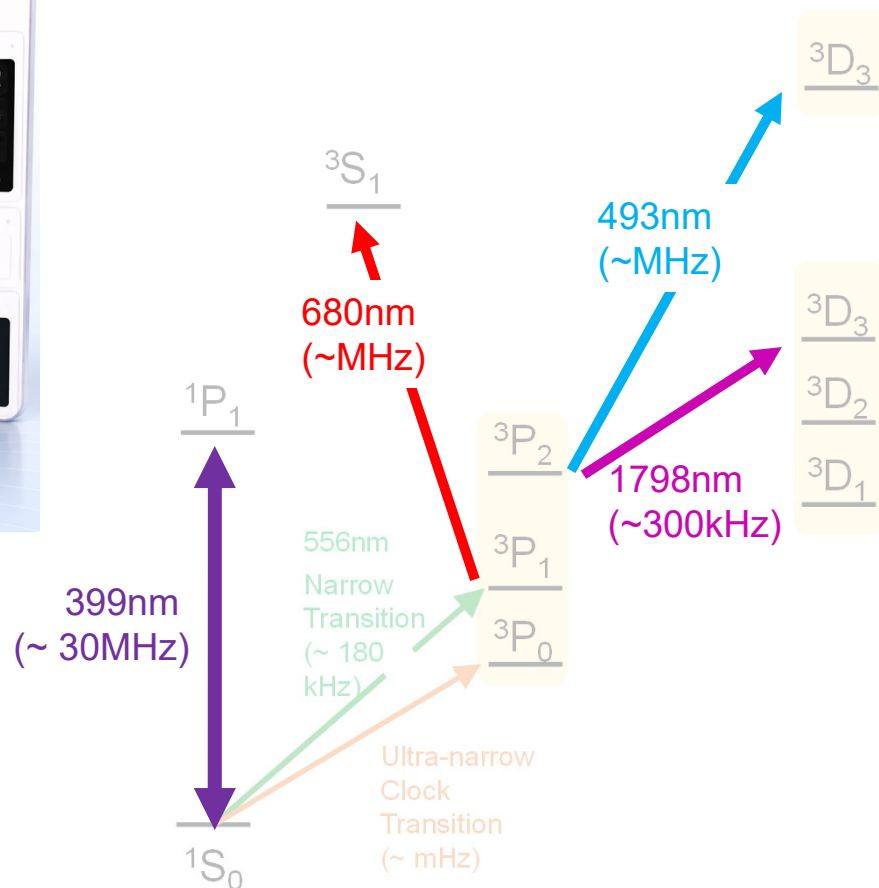
Science: figuring out qubit control schemes



中央研究院
ACADEMIA SINICA



Quantum engineer ready ??



Hao-Rong Yang 楊皓蓉
Master thesis (in preparation)